

The $6N$ -Skeleton Zeta Function and its Log-Periodic Zero-Ladders

Ruqing Chen

GUT Geoservice Inc., Montreal, Quebec, Canada

ruqing@hotmail.com

June 8, 2026

Abstract

The $6N$ skeleton is built by stripping the primes 2 and 3, so that every remaining prime sits on a fault line $6N \pm 1$. We make that stripping explicit at the level of the zeta function. Removing the $q = 2$ and $q = 3$ Euler factors defines $\zeta_{6N}(s) = \zeta(s)(1 - 2^{-s})(1 - 3^{-s}) = \prod_{p \geq 5} (1 - p^{-s})^{-1} = \sum_{\gcd(n,6)=1} n^{-s}$, the Dirichlet series of the integers coprime to 6. Its analytic structure is elementary but clean: a simple pole at $s = 1$ with residue $1/3$, and a zero set that splits into two orthogonal families—the non-trivial zeros of ζ on the critical line $\beta = \frac{1}{2}$, unchanged, and two *new* ladders of zeros on the imaginary axis $\beta = 0$, at $s = 2\pi ik/\ln 2$ and $s = 2\pi in/\ln 3$. Because $\ln 2/\ln 3$ is irrational the two ladders are incommensurate, meeting only at the double zero $s = 0$. We show these ladders are exactly the Fourier modes of the two log-periodic staircases ψ_2, ψ_3 that count the powers of 2 and 3 removed from $\psi(x)$; equivalently, the skeleton's prime count $\psi_{6N}(x) = \sum_{p \geq 5, p^k \leq x} \ln p$ decomposes into a smooth term, the fluctuating critical-line ($\beta = \frac{1}{2}$) part, and a rigid, exactly log-periodic ($\beta = 0$) grid. We are explicit about scope: ζ_{6N} is ζ with two Euler factors removed, its extra zeros are a construction artifact of that removal, and nothing here bears on the Riemann Hypothesis or on the infinitude of any prime constellation.

1 Introduction

Throughout this series the $6N$ skeleton has been defined operationally: discard the primes 2 and 3, and every surviving prime (> 3) lies at $6N \pm 1$. Volumes I–II worked with this skeleton combinatorially and in congruence phase space. Here we ask the obvious complex-analytic question: what is the Dirichlet generating function of the skeleton, and what does “removing 2 and 3” do to the zeros of ζ ?

The answer is modest and exact. Removing the two Euler factors multiplies ζ by $(1 - 2^{-s})(1 - 3^{-s})$, which introduces two arithmetic-progression ladders of zeros on the imaginary axis. These are not deep new zeros—they are the spectral fingerprint of the two log-periodic staircases (the powers of 2 and of 3) that the skeleton throws away. What is mildly satisfying is the clean separation that results: a rigid periodic grid at $\beta = 0$ and the familiar fluctuating zeros at $\beta = \frac{1}{2}$, sitting side by side in one generating function.

2 Motivation: where the skeleton's prime energy lives

Before defining ζ_{6N} we record the elementary fact that motivates removing 2 and 3. The von Mangoldt weight $\Lambda(n) = \ln p$ for $n = p^k$ is supported on prime powers; for $p \geq 5$ these sit at $n \equiv \pm 1 \pmod{6}$, while the powers of 2 and 3 occupy the residues $\{2, 4\}$ and $\{3\}$, and no prime power is $\equiv 0$. Binning $\sum_{n \leq X} \Lambda(n)$ by residue mod 6 at $X = 10^7$ (Fig. 1) shows 99.9997% of the total weight on the fault lines $6N \pm 1$, split essentially evenly, with only $O(\log X)$ left on the powers of 2, 3 and exactly nothing on the multiples of 6.

Residue-resolved von Mangoldt energy on the 6N skeleton (X=10⁷): a check recovering classical facts

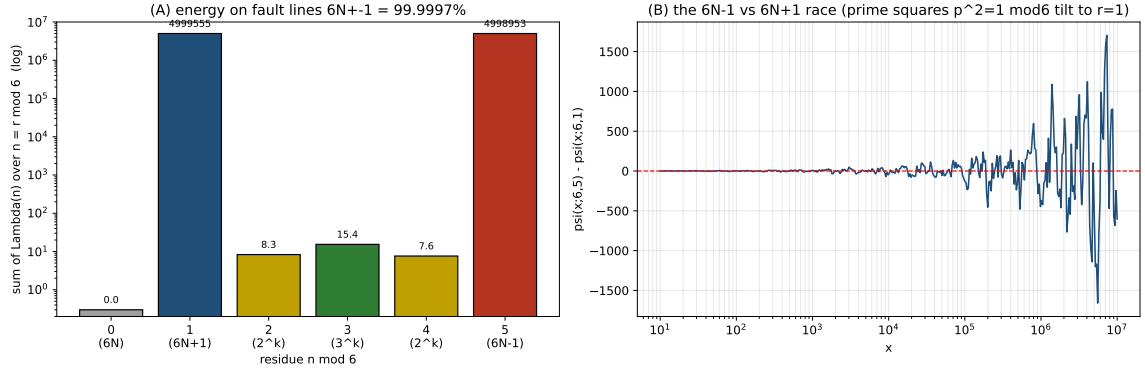


Figure 1: (A) The von Mangoldt energy $\sum_{n \leq X, n \equiv r} \Lambda(n)$ by residue $r \pmod 6$ at $X = 10^7$: the fault lines $6N \pm 1$ carry 99.9997%, the residues 2, 3, 4 only $O(\log X)$ (powers of 2, 3), and $r = 0$ nothing. (B) The $6N-1$ vs $6N+1$ race $\psi(x; 6, 5) - \psi(x; 6, 1)$, oscillating at scale $O(\sqrt{x})$. These are the classical facts—prime distribution in arithmetic progressions and its $\sqrt{x}$ fluctuation—displayed on the skeleton; they are not new, and we use them only to justify isolating the $6N \pm 1$ generators.

This is exactly the prime number theorem in the arithmetic progressions $(\pmod 6)$ [2], with the $\sqrt{x}$ -scale race a face of the explicit formula. We claim no novelty here; it is the empirical reason to define a generating function that keeps only the $p \geq 5$ factors.

3 The $6N$ -skeleton zeta function

Definition 1. For $\text{Re}(s) > 1$,

$$\zeta_{6N}(s) := \zeta(s)(1 - 2^{-s})(1 - 3^{-s}) = \prod_{p \geq 5} (1 - p^{-s})^{-1} = \sum_{\substack{n \geq 1 \\ \gcd(n, 6) = 1}} n^{-s}, \quad (1)$$

the Dirichlet series of the integers coprime to 6 (equivalently, the 5-rough integers). It continues meromorphically to $\mathbb{C}$ through ζ .

Proposition 1 (Pole). ζ_{6N} has a single simple pole, at $s = 1$, with residue $(1 - \frac{1}{2})(1 - \frac{1}{3}) = \frac{1}{3}$, reflecting the density $\varphi(6)/6 = 1/3$ of the integers coprime to 6.

Proof. ζ has a simple pole at $s = 1$ with residue 1; the analytic factor $(1 - 2^{-s})(1 - 3^{-s})$ takes the value $\frac{1}{2} \cdot \frac{2}{3} = \frac{1}{3}$ there and is holomorphic, so the residue is $1 \cdot \frac{1}{3}$. $\square$

4 The zero set: a critical-line family and two imaginary ladders

Since $\zeta_{6N} = \zeta \cdot (1 - 2^{-s})(1 - 3^{-s})$, its zeros are the union of the zeros of the three factors.

Proposition 2 (Zero set). The zeros of ζ_{6N} are:

1. the non-trivial zeros $\rho = \frac{1}{2} + i\gamma$ of ζ (unchanged), together with the trivial zeros $s = -2, -4, \dots$;
2. the ladder $L_2 = \{2\pi i k / \ln 2 : k \in \mathbb{Z}\}$, from $1 - 2^{-s} = 0$;
3. the ladder $L_3 = \{2\pi i n / \ln 3 : n \in \mathbb{Z}\}$, from $1 - 3^{-s} = 0$.

Both ladders lie on the imaginary axis $\beta = 0$, with constant spacings $2\pi/\ln 2 \approx 9.0647$ and $2\pi/\ln 3 \approx 5.7192$. Since $\ln 2/\ln 3$ is irrational, $L_2 \cap L_3 = \{0\}$, where the two simple zeros coincide into a double zero of ζ_{6N} .

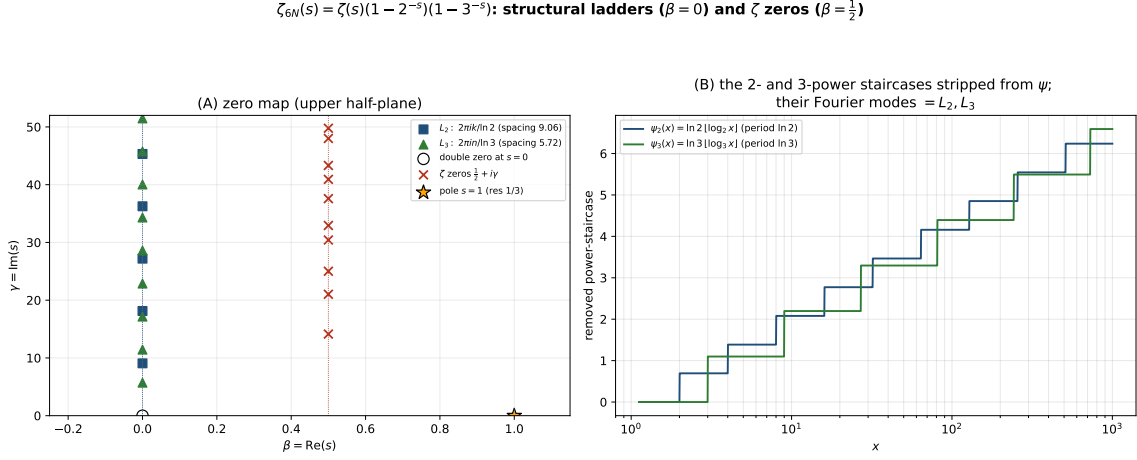


Figure 2: (A) Zero map of ζ_{6N} (upper half-plane): the incommensurate imaginary ladders L_2 (spacing 9.06) and L_3 (spacing 5.72) at $\beta = 0$, the double zero at $s = 0$, the critical-line zeros $\frac{1}{2} + i\gamma$ of ζ , and the pole at $s = 1$. (B) The log-periodic staircases $\psi_2(x) = \ln 2[\log_2 x]$ and $\psi_3(x) = \ln 3[\log_3 x]$ removed from ψ ; their Fourier frequencies are precisely L_2 and L_3 .

5 The ladders are the Fourier modes of the removed staircases

The two new families are not arithmetic accidents; they are the spectrum of what the skeleton discards. The skeleton's prime count is

$$\psi_{6N}(x) = \sum_{\substack{p \geq 5 \\ p^k \leq x}} \ln p = \psi(x) - \psi_2(x) - \psi_3(x), \quad \psi_2(x) = \ln 2 \left\lfloor \frac{\ln x}{\ln 2} \right\rfloor, \quad \psi_3(x) = \ln 3 \left\lfloor \frac{\ln x}{\ln 3} \right\rfloor, \quad (2)$$

where ψ_2, ψ_3 are staircases of period $\ln 2, \ln 3$ in $\ln x$. Their contribution to the logarithmic derivative is

$$-\frac{d}{ds} \ln(1 - 2^{-s}) = -\ln 2 \sum_{k \geq 1} (2^k)^{-s}, \quad (3)$$

the Dirichlet series supported on powers of 2; the zeros L_2 of $1 - 2^{-s}$ are exactly the frequencies of the Fourier expansion of the sawtooth $\{\log_2 x\}$, and likewise L_3 for 3. Reading ψ_{6N} through the explicit formula therefore gives a clean three-way split:

$$\psi_{6N}(x) = \underbrace{x}_{\text{smooth}} - \underbrace{\sum_{\substack{\rho \\ \beta = \frac{1}{2} \text{ fluctuation}}} \frac{x^\rho}{\rho}}_{\beta = \frac{1}{2} \text{ fluctuation}} - \underbrace{(\psi_2(x) + \psi_3(x))}_{\beta=0 \text{ log-periodic grid}} + (\text{lower order}). \quad (4)$$

The $\beta = 0$ ladders form a rigid, exactly periodic grid (the discarded 2, 3 residues); the $\beta = \frac{1}{2}$ zeros carry the fluctuation. In this precise sense the structural and the fluctuating parts of the skeleton's prime count are orthogonal: one is a finite-rate set of pure log-frequencies, the other the critical-line spectrum.

6 Conclusion

The $6N$ -skeleton zeta function $\zeta_{6N}(s) = \zeta(s)(1 - 2^{-s})(1 - 3^{-s})$ is the generating function of the integers coprime to 6. Its only departures from ζ are bookkeeping consequences of removing the 2 and 3 Euler factors: the residue at $s = 1$ shrinks to $1/3$, and two incommensurate ladders of zeros appear on the imaginary axis, encoding the log-periodic staircases of powers of 2 and 3. The resulting separation—a rigid $\beta = 0$ grid beside the fluctuating $\beta = \frac{1}{2}$ zeros—is a tidy way to see what the skeleton keeps and what it throws away. We stress the limits of the claim: this is an elementary modification of ζ , the new zeros are construction artifacts rather than discoveries, and the framework says nothing about the Riemann Hypothesis or the infinitude of primes or prime constellations beyond what classical theory already gives.

Data and code availability. The residue-energy computation, the zero-map and staircase figure, and a verifier of ζ_{6N} 's pole, ladders, and incommensurability are openly available at <https://github.com/Ruqing1963/6N-skeleton-zeta> [7].

References

- [1] B. Riemann, *Ueber die Anzahl der Primzahlen unter einer gegebenen Grösse*, Monatsber. Berlin Akad. (1859).
- [2] H. Davenport, *Multiplicative Number Theory*, 3rd ed., Springer GTM 74 (2000).
- [3] R. Chen, Part V, Zenodo, doi:10.5281/zenodo.20510700 (2026).
- [4] R. Chen, *The circular convolution of prime constellations in congruence phase space* (Part XX), Zenodo, doi:10.5281/zenodo.20583994 (2026).
- [5] R. Chen, *The quasi-periodic diffraction spectrum of twin-pair correlation on the $6N$ skeleton* (Part XXIV), Zenodo, doi:10.5281/zenodo.20587185 (2026).
- [6] R. Chen, *Arithmetic geodynamics on the $6N$ skeleton: a review of Parts I–XIX*, Zenodo, doi:10.5281/zenodo.20585301 (2026).
- [7] R. Chen, *The $6N$ -skeleton zeta function and its log-periodic zero-ladders* (Part XXV), <https://github.com/Ruqing1963/6N-skeleton-zeta> (2026).